

Perovskite Encapsulant Comparison | Eversolar Series

Customer-facing summary based on reported WVTR / permeability data. Lower permeability and WVTR indicate stronger moisture barrier performance.

Best moisture barrier	Fast low-temperature process	Application fit
Eversolar AB-341	Eversolar series	AB-341 + AB-302
0.0074 gmm/m2day @1 mm Approx. 32x lower permeability than the nearest PIB benchmark in this table. UV/VIS 365-405 nm, 10 sec Room-temperature curing avoids long thermal lamination cycles. AB-341 for strongest barrier; AB-302 where transparent adhesive is preferred.		

Priority	Product	Supplier / Product Group	Positioning	Curing Process	Transparency	Application	Permeability @1mm	Relative Barrier Index
Eversolar	Everlight Eversolar® AB-341	Eversolar Series	Best barrier sealing	365~405nm;10 sec. Room temp.	White	Carpet Sealing Frame/Edge sealing	0.0074	1.000
Eversolar	Everlight Eversolar® AB-302	Eversolar Series	Transparent fast-curing adhesive option	365~405nm;10 sec. Room temp.	Transparent	Carpet Sealing	0.8000	0.009
Benchmark	H.B. Fuller HelioSeal® PVS-101	PIB edge seal competitor	External benchmark	110~157°C. 10~15 min	Black	Frame/Edge sealing	0.2340	0.032
Benchmark	Quanex SET LP03	PIB edge seal competitor	External benchmark	110~157°C. 10~15 min	Black	Frame/Edge sealing	0.4560	0.016
Benchmark	Hangzhou First POE (TF Series)	Thermal film competitor	External benchmark	140~155°C 12~20 min.	Transparent	Carpet Sealing	2.0200	0.004
Benchmark	Hangzhou First TPO (TP8-NSX)	Thermal film competitor	External benchmark	140~160°C ?min.	Transparent	Carpet Sealing	2.0200	0.004
Benchmark	3M™ PO8110	Thermal film competitor	External benchmark	150°C for 15 min	Transparent	Carpet Sealing	2.6000	0.003
Benchmark	DELO PHOTOBOND LP4224	UV adhesive competitor	External benchmark	365~405nm;10 sec. Room temp.	Transparent	Carpet Sealing	10.0000	0.001
Benchmark	EVA Film	Thermal film competitor	External benchmark	140~155°C 12~20 min.	Transparent	Carpet Sealing	13.5000	0.001

Visual takeaway: lower permeability indicates stronger moisture barrier. Eversolar AB-341 is the reference leader in this comparison.

Product	Permeability @1mm
Eversolar® AB-341	0.0074
H.B. Fuller HelioSeal® PVS-101	0.2340
Quanex SET LP03	0.4560
Eversolar® AB-302	0.8000
Hangzhou First POE (TF Series)	2.0200
Hangzhou First TPO (TP8-NSX)	2.0200
3M™ PO8110	2.6000
DELO PHOTOBOND LP4224	10.0000
EVA Film	13.5000

